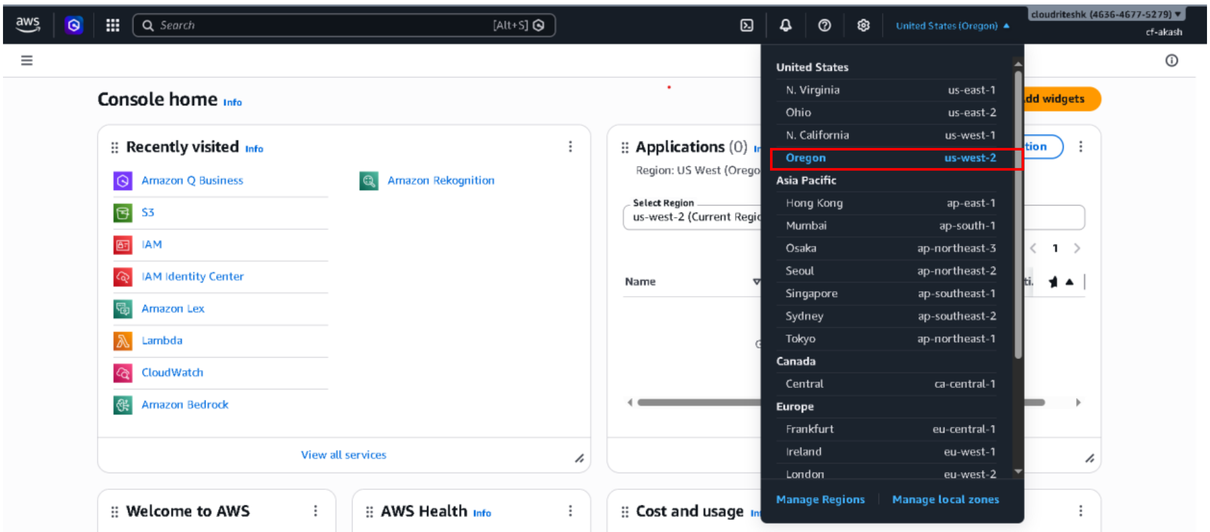
Amazon Bedrock Knowledge Base: -

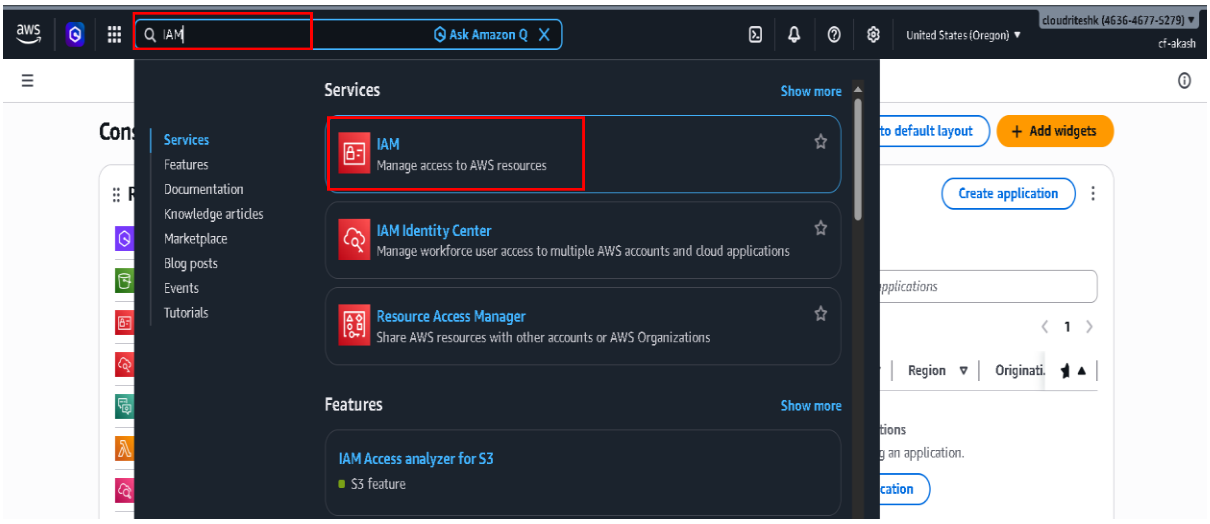
Summary: -

Amazon Bedrock Knowledge Base allows users to create an AI-powered system by connecting data sources like Amazon S3 and using vector stores for intelligent retrieval. The process involves creating an IAM user (since root access is restricted), setting up S3 buckets to store data, and configuring a knowledge base in Amazon Bedrock. Users select a model, define data sources, and use OpenSearch Serverless for vector storage. After syncing the data, the system can answer user queries based on the uploaded documents, providing accurate and source-linked responses.

- Before you start please shift to region us-west-2.
- Because some of the services not work in other regions.

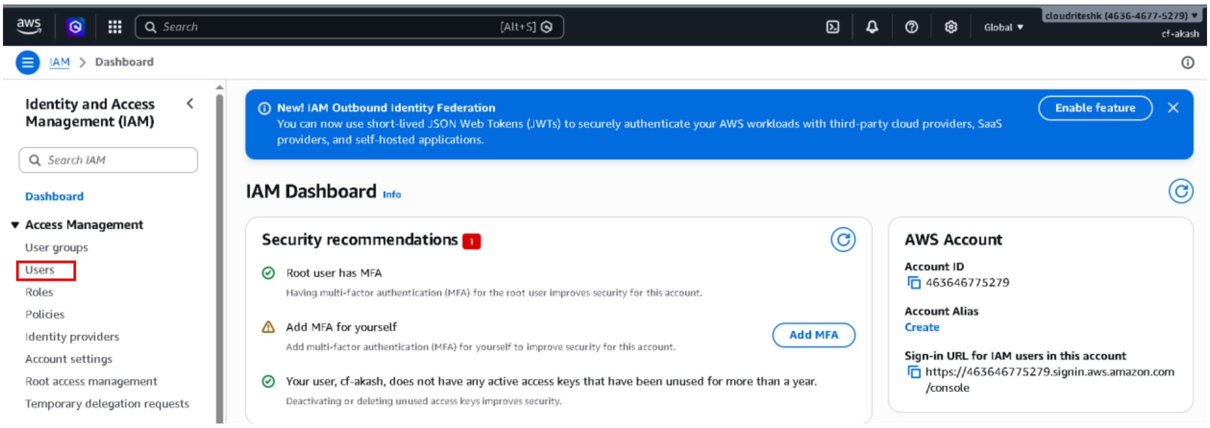


- Second thing you can't create Amazon Bedrock Knowledge Base if you logged in as root user

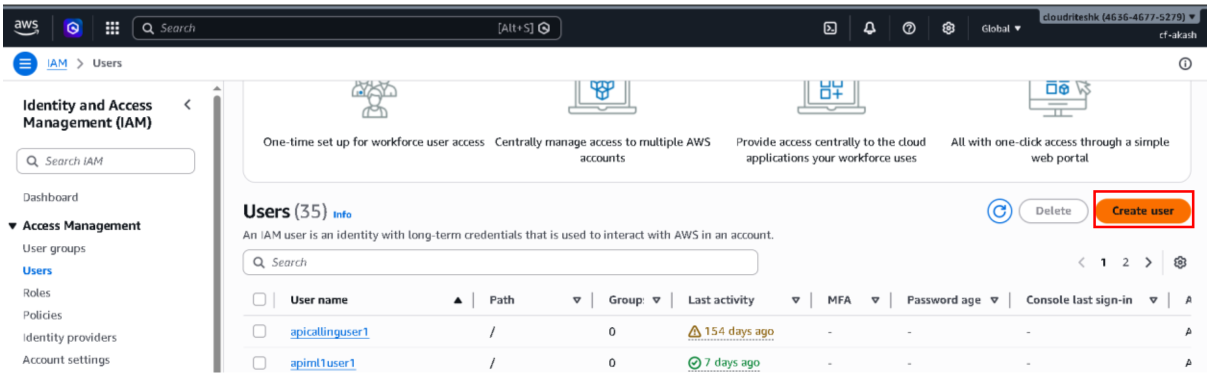


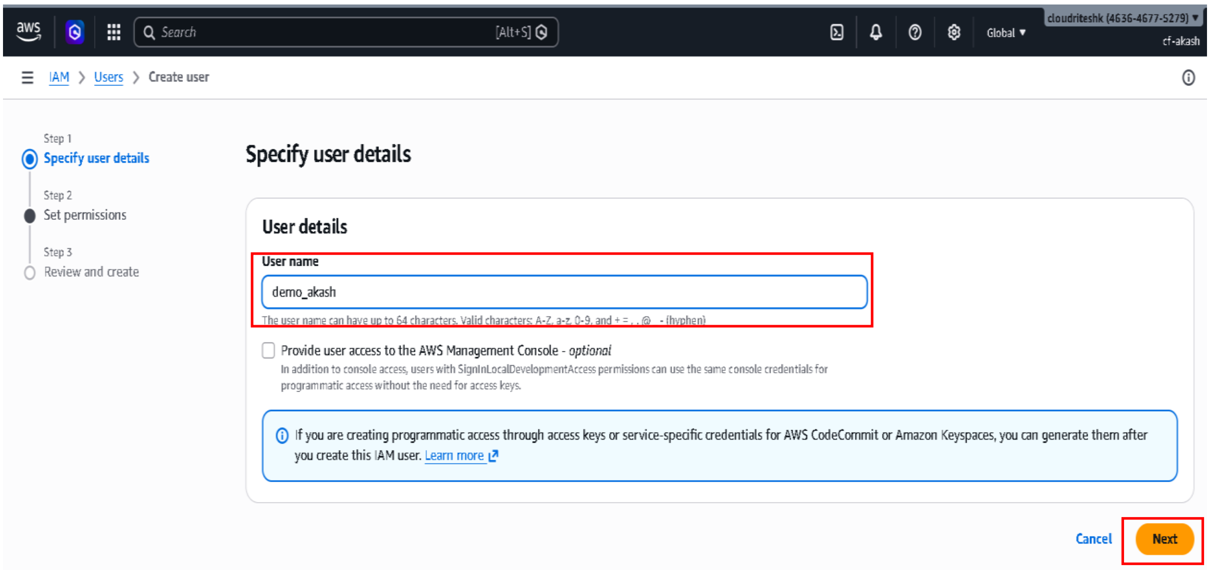
- So what you will have to do is you will have to create an IAM user.
- How you create IAM user: -
    - ➢ Go and search IAM in AWS console Search bar

- Now click on IAM

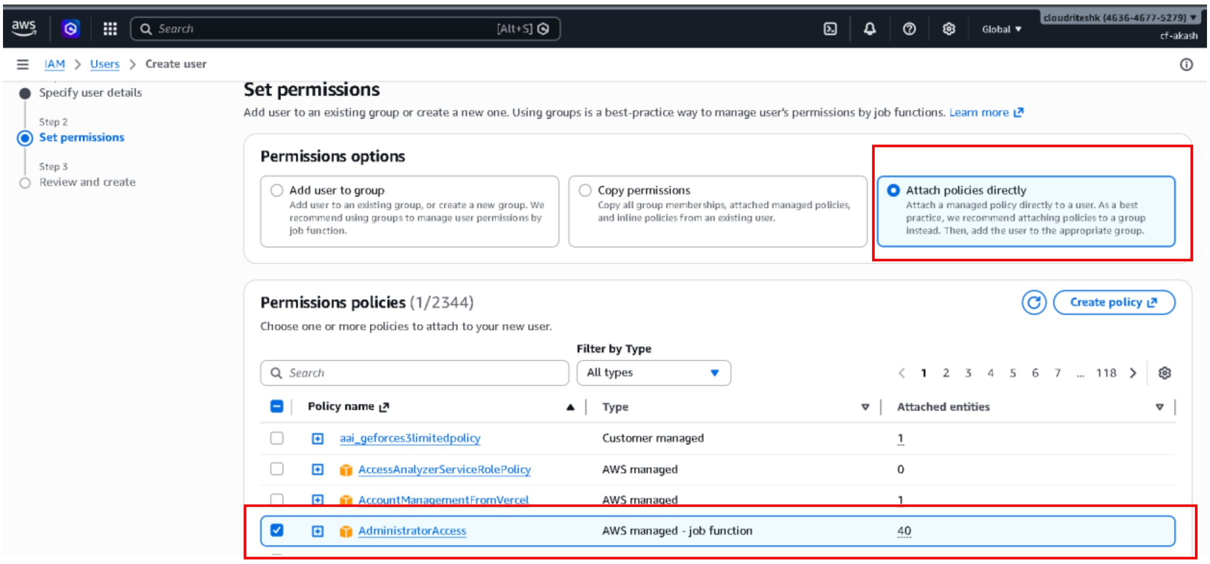


- Now click on user

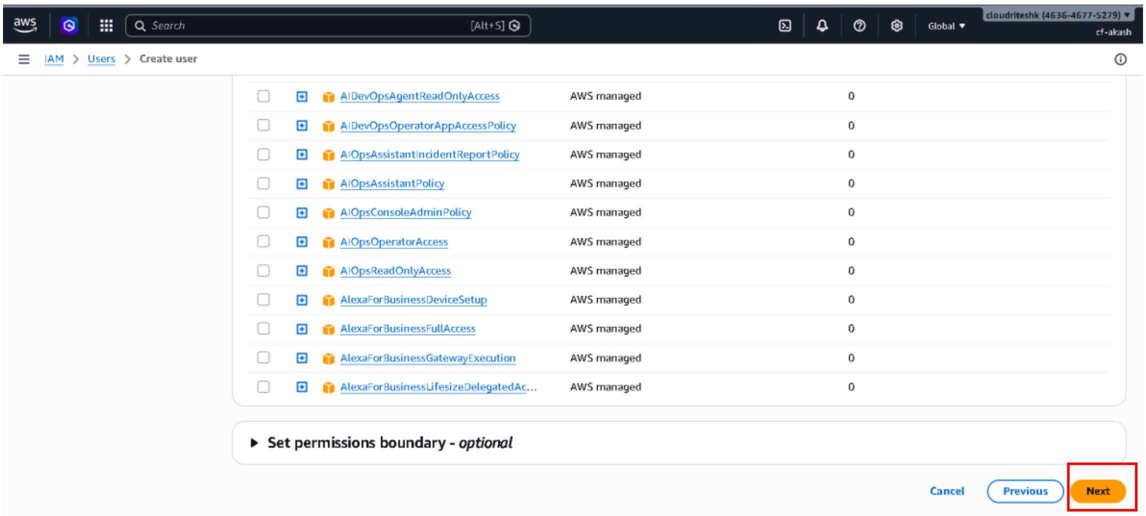


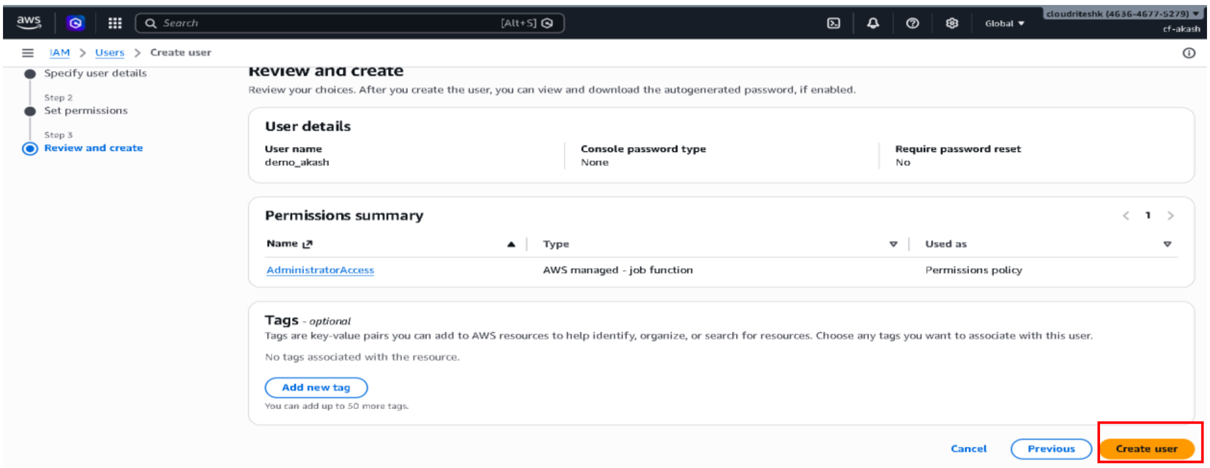


- Now click on create user
- Enter User name and Enter Next.

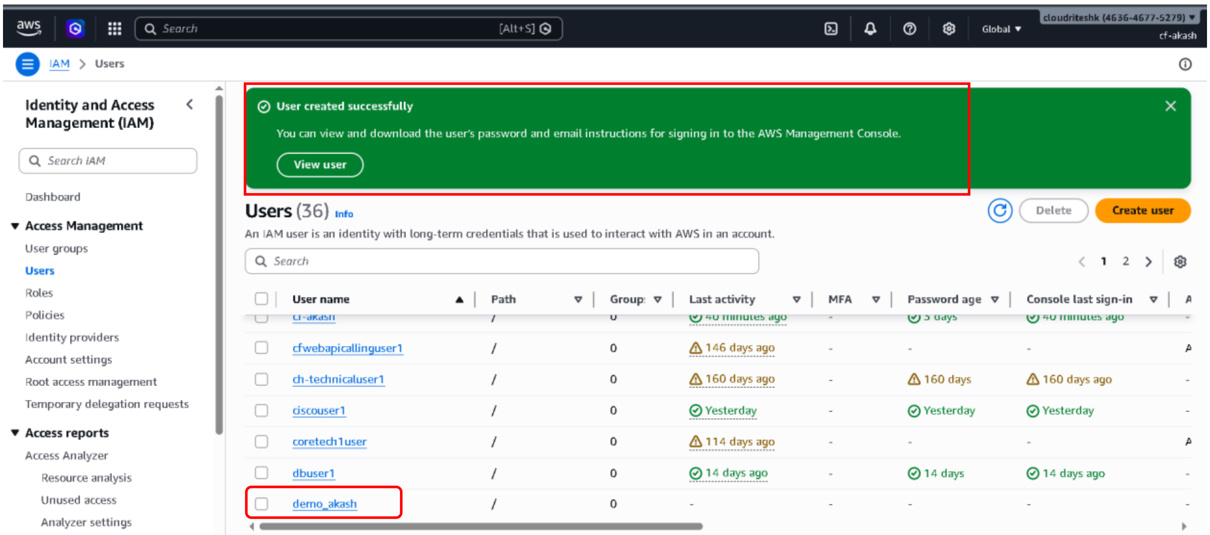


- Select Permission options: -
  - ➢ Attach policies directly
- Add Permission policies: -
  - ➢ AdministratorAccess
- Now just click Next

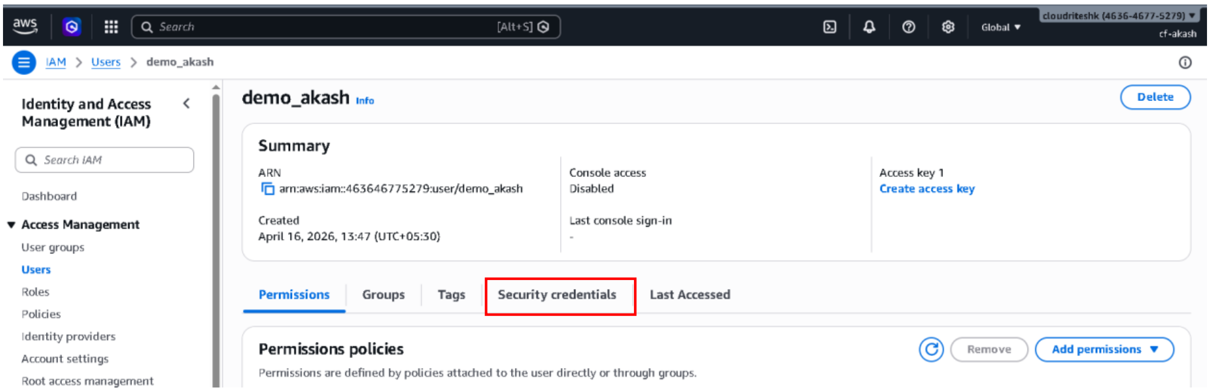




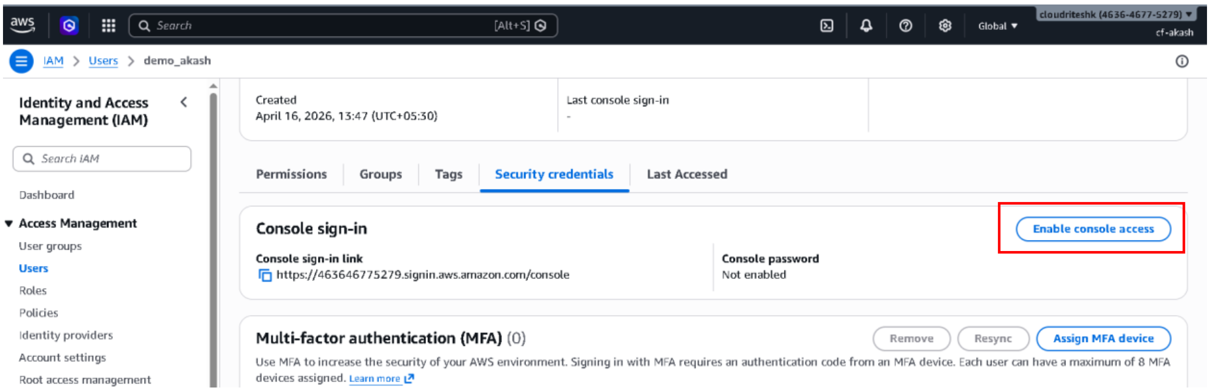
- Click create user.



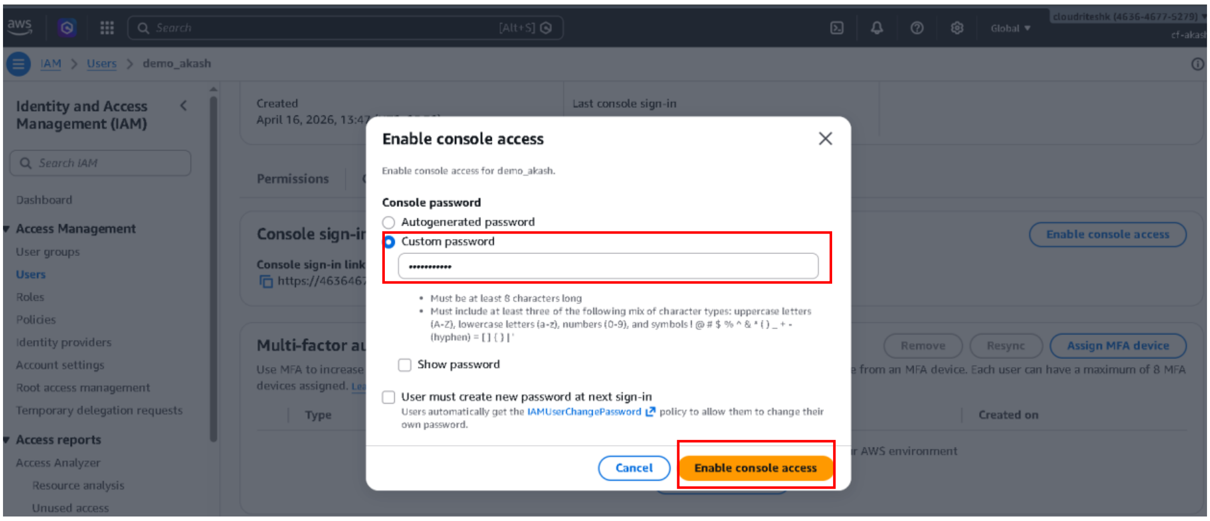
- Now our IAM user is created
- Click on the created user and go inside



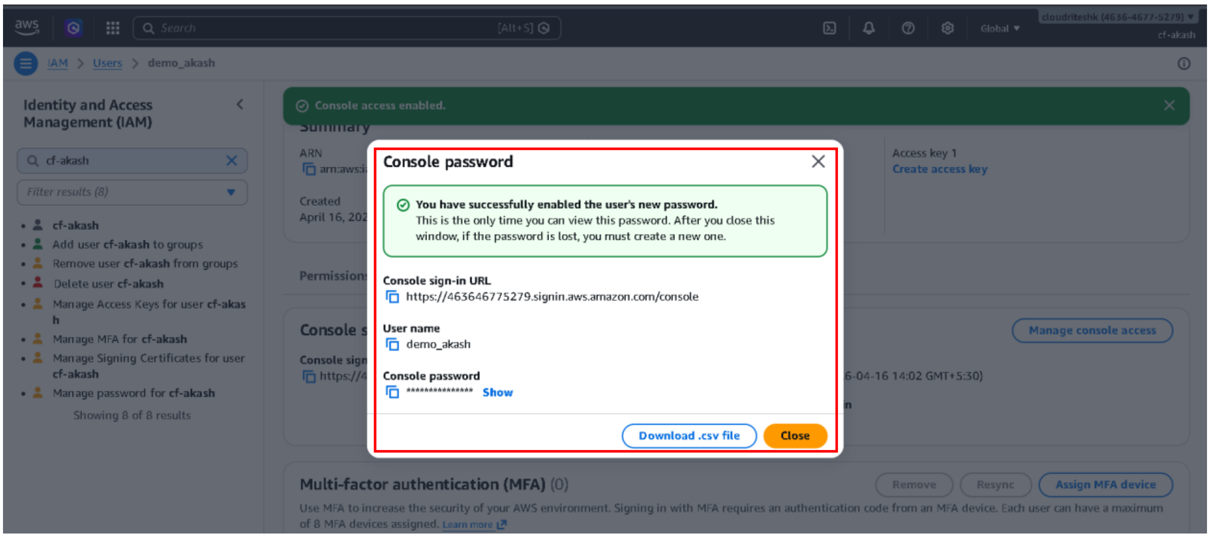
- Click on the security credentials.



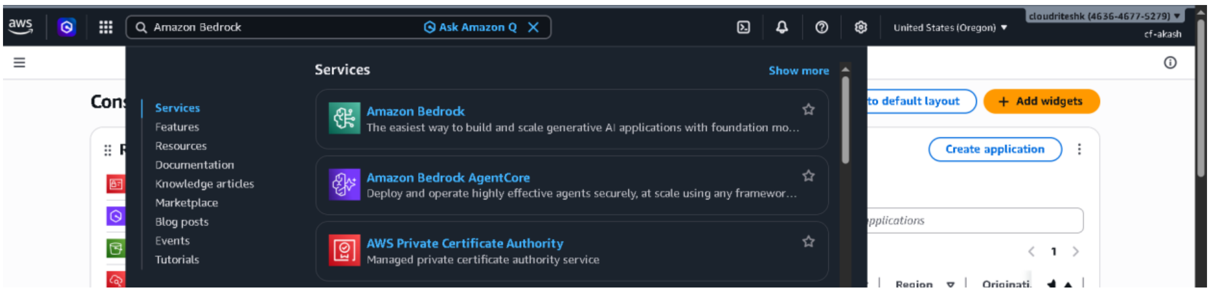
- Now just click Enable console access



- Now just Enter your password and click Enable console access

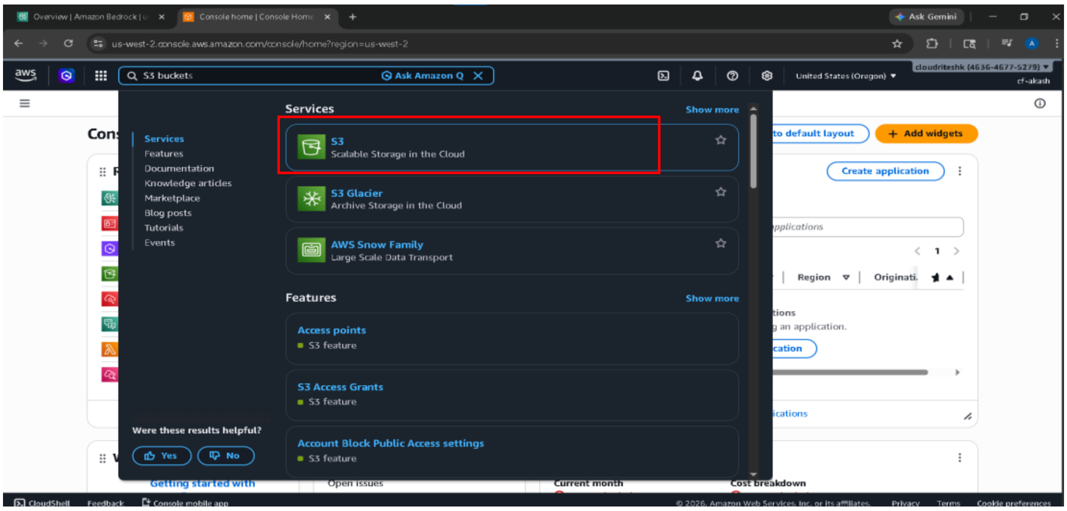


- Now just Sign out from root user and login from this IAM user that you have created.
- so now let’s go and create our Amazon Bedrock Knowledge Base.

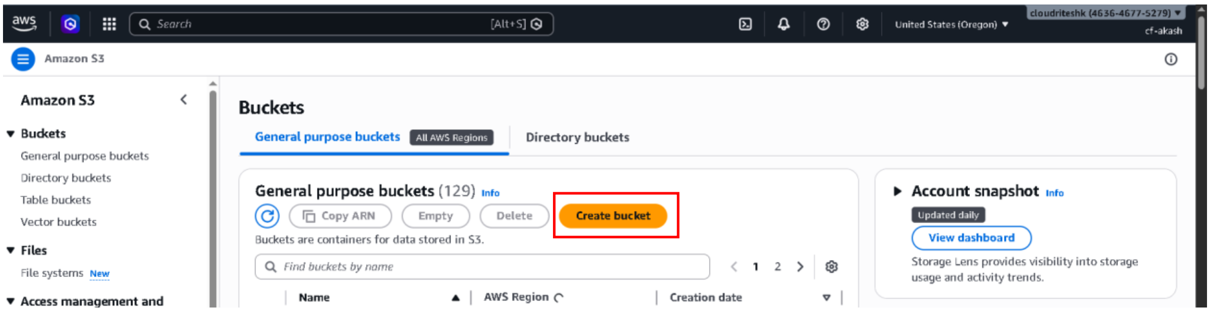


- Got to AWS console and search Amazon Bedrock.
- Click on Amazon Bedrock.

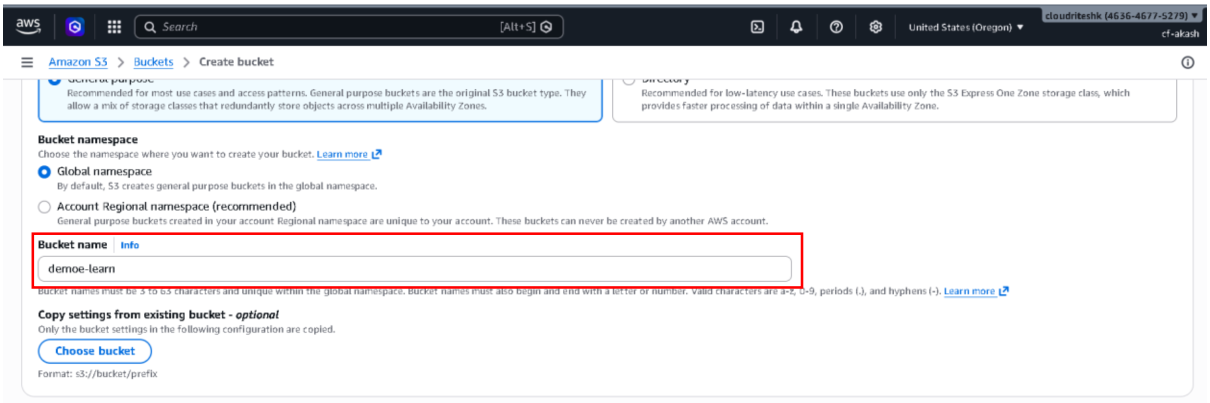
- The First thing for our E-learning application is
- Identifying Data source
- So we have to create an S3 bucket and upload all relative files which we want the user to be able to access as part of this E-learning platform.
- So let’s create S3 Bucket first



- Open Aws in new tab and Search S3 bucket
- Click on S3

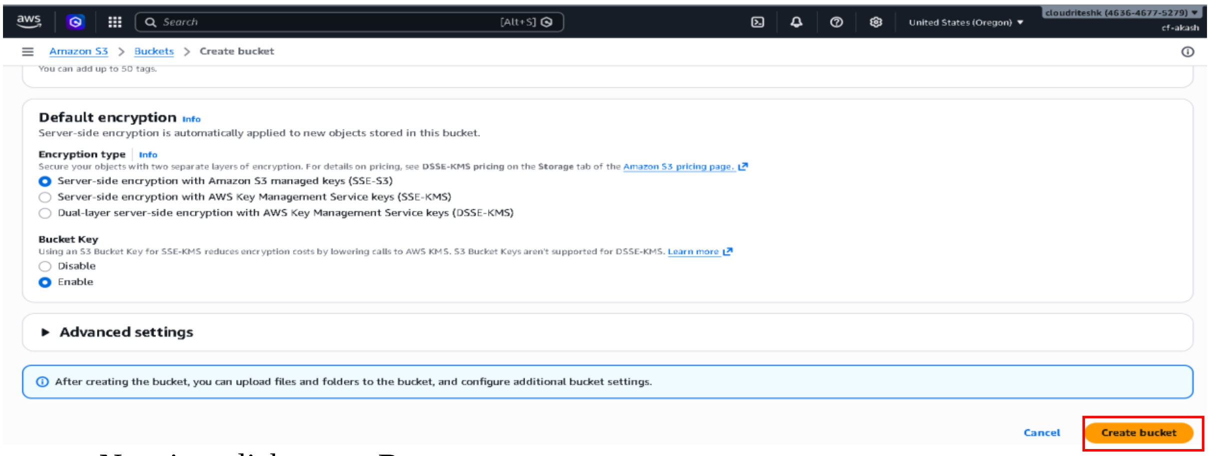


- Click Create bucket

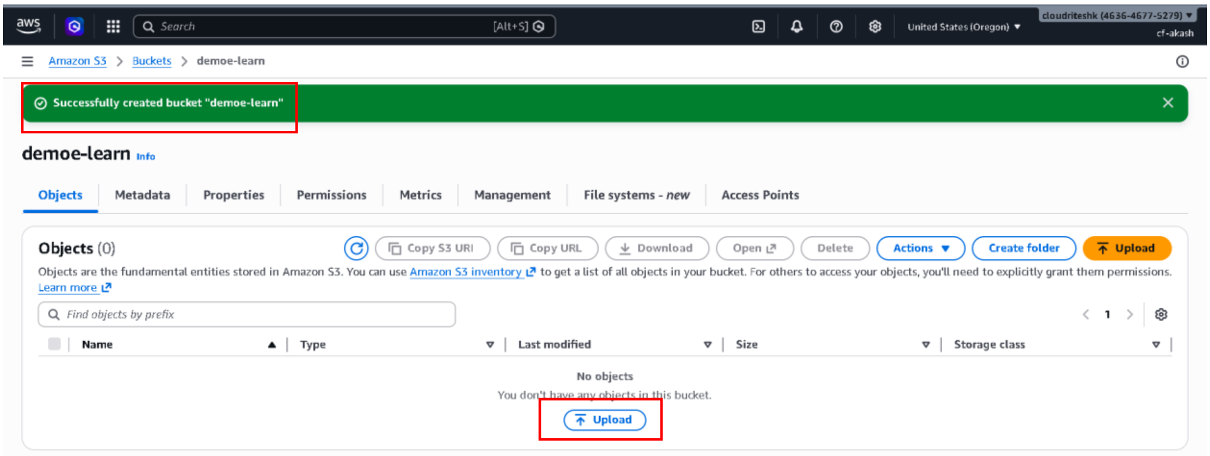


- Enter Bucket name.

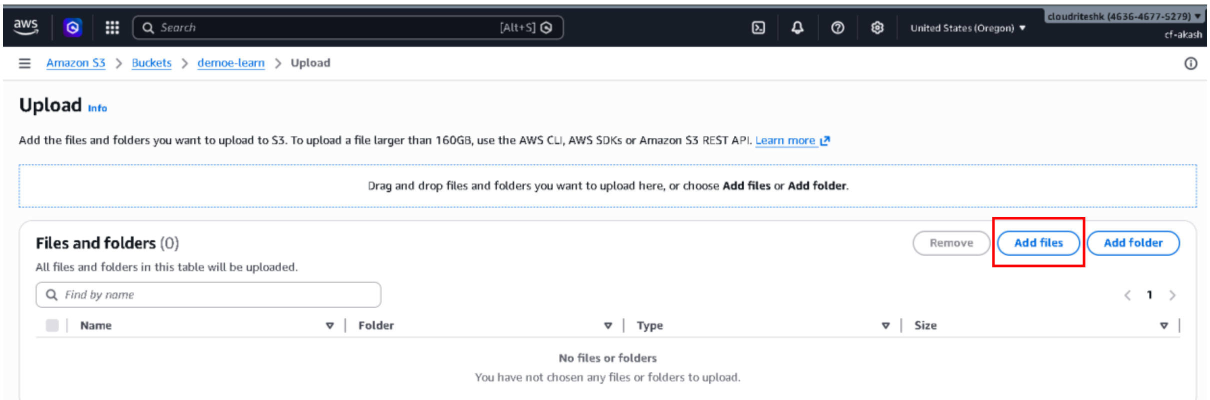
- Keep everything else default.



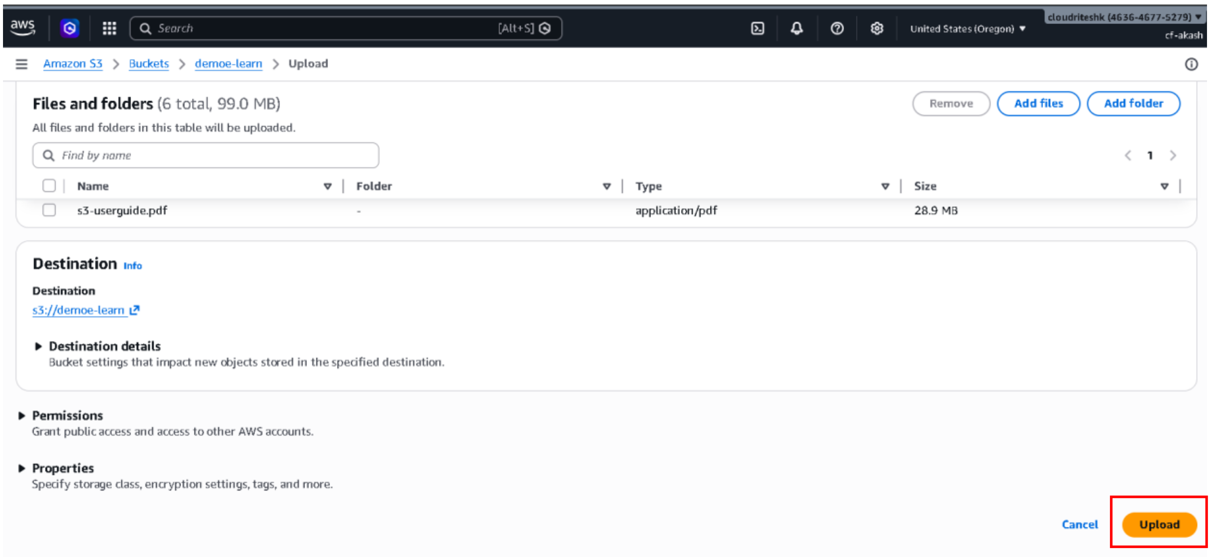
- Now just click create Button.



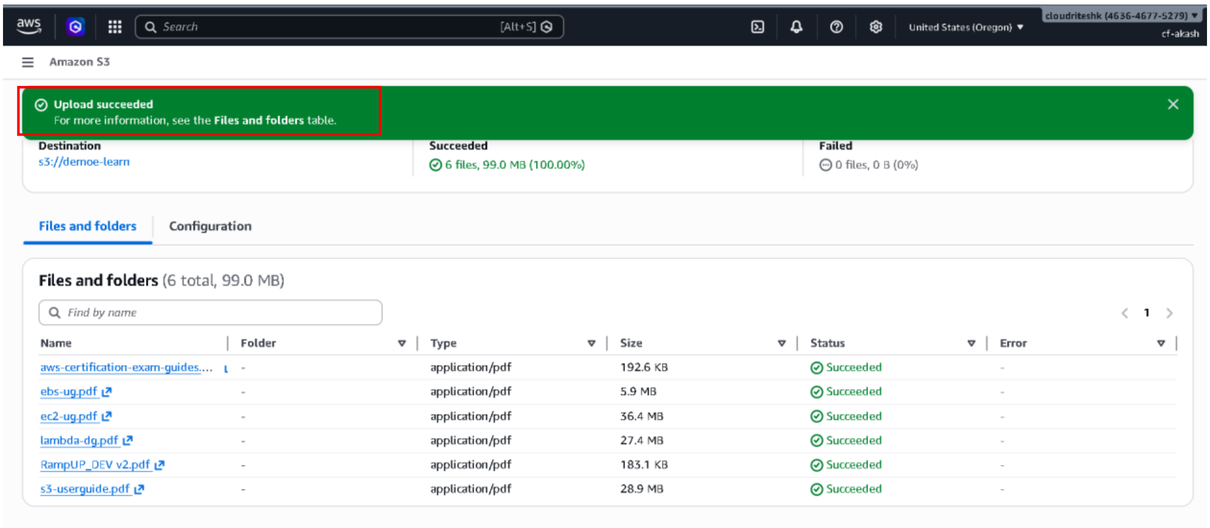
- Our S3 bucket is created.
- Now click on upload.



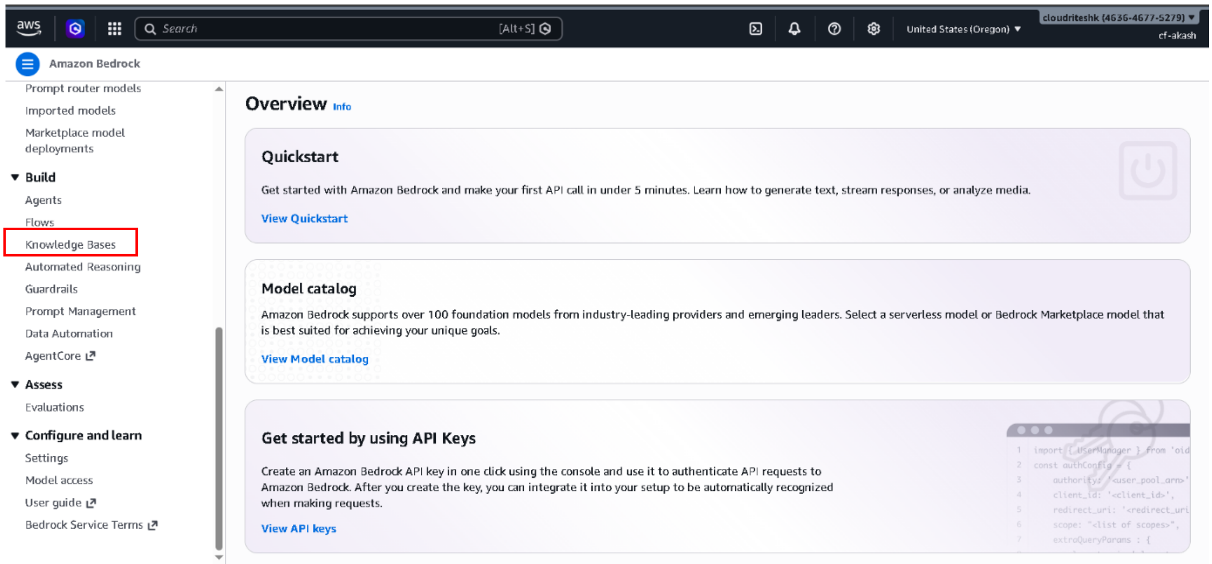
- Click Add files.
- And select files which we will use our internal data source.



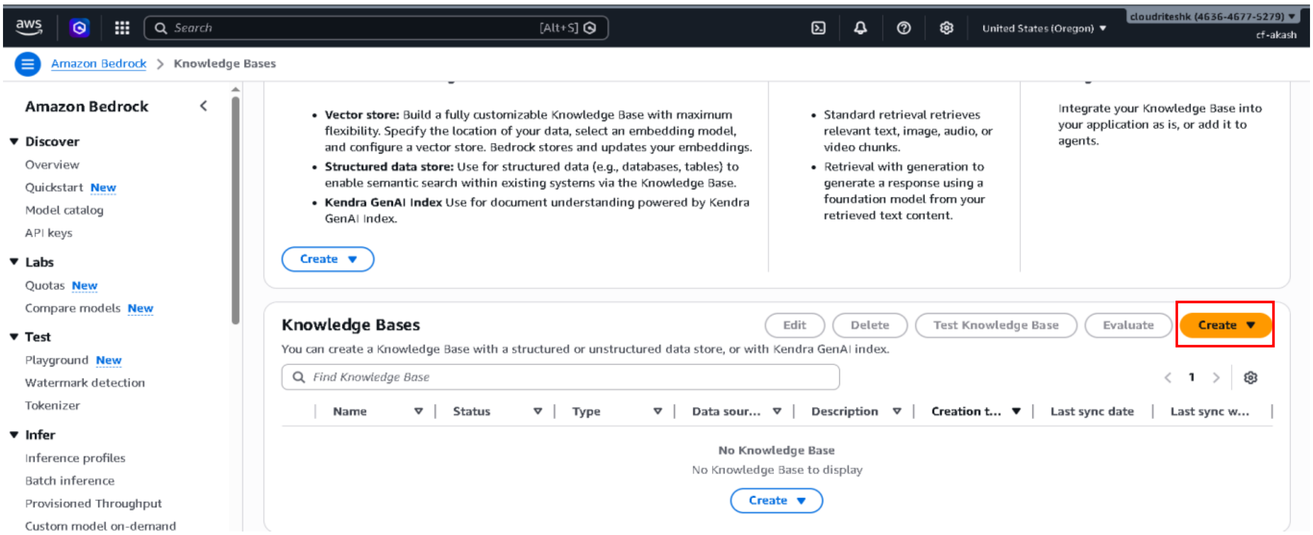
- Now just click upload



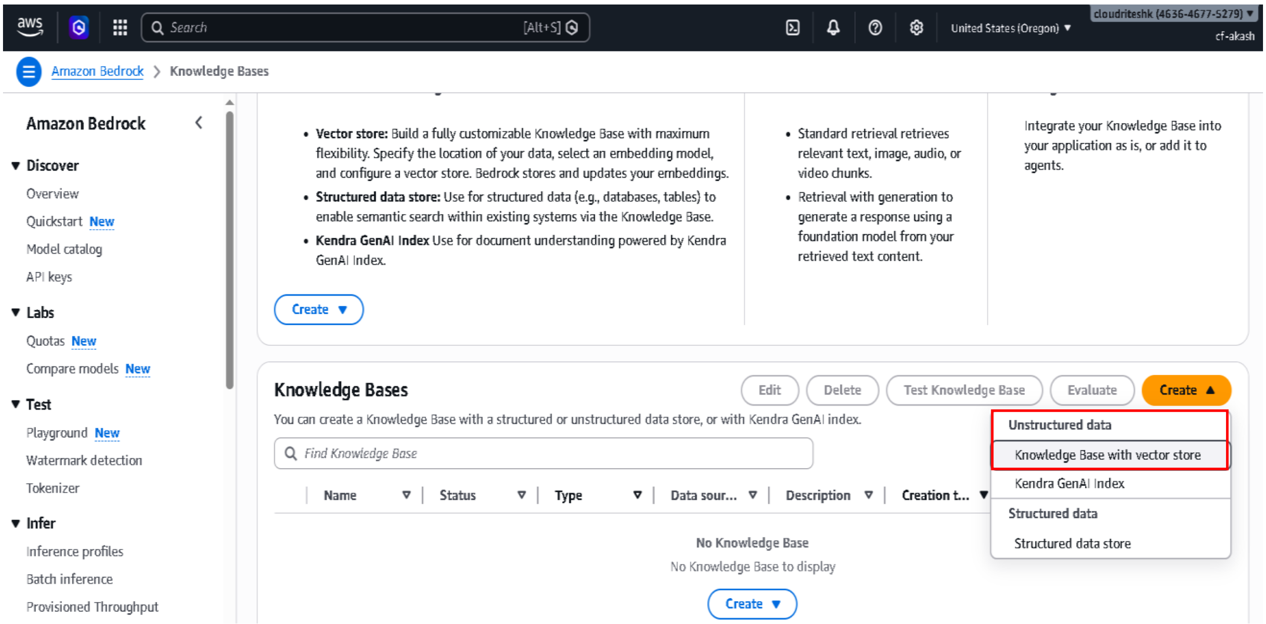
- Now our all Files is uploaded.
- Basically these files are AWS Documents like: -Amazon Ec2 guide, Lambda guide etc.
- Now Go to your Amazon Bedrock tab.



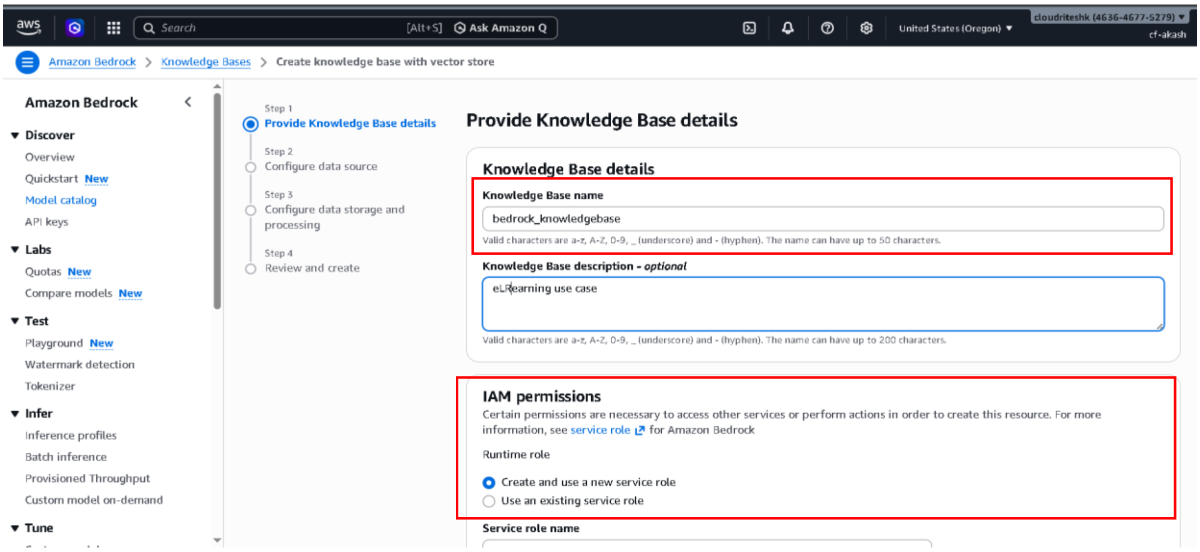
- Scroll down in navigation bar.
- Inside Build click knowledge Bases
- Before creating knowledge base I just want to warn you that’s there going to be some cost involved.
- Because we are going to create a AWS open search server less vector store and that cost about $0.50 per hour.



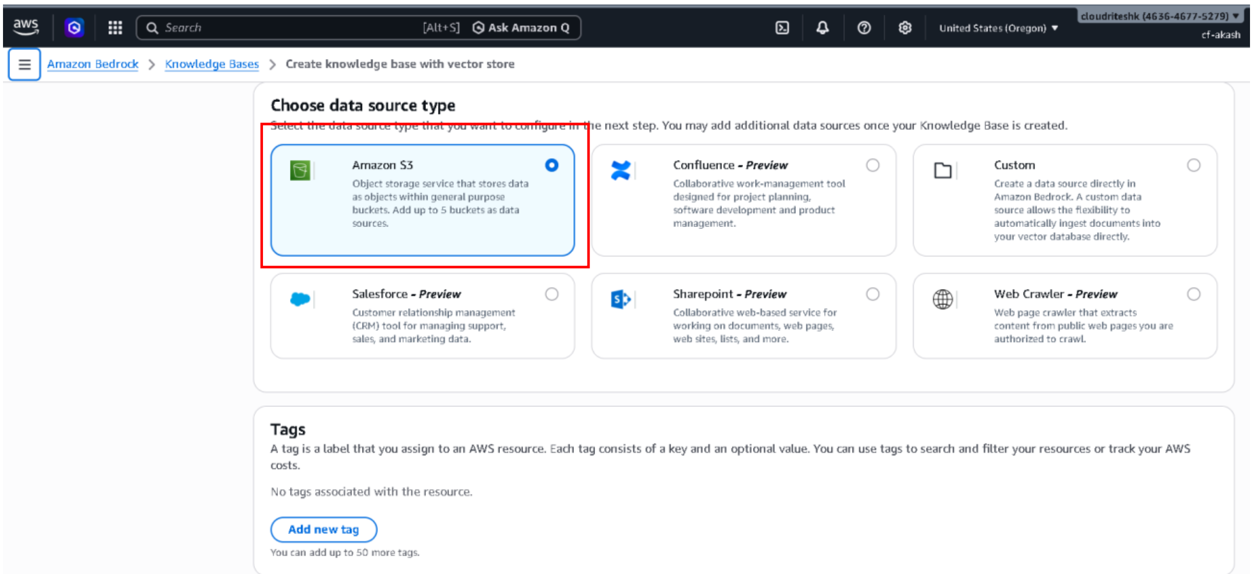
- Now just click on create.



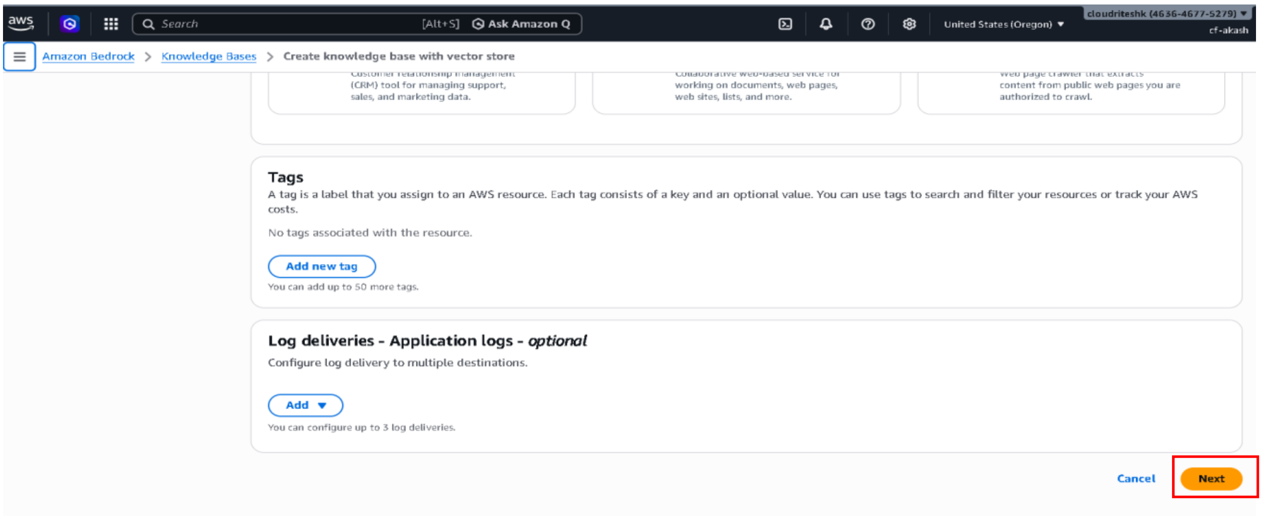
- Choose option Knowledge Base with vector store.



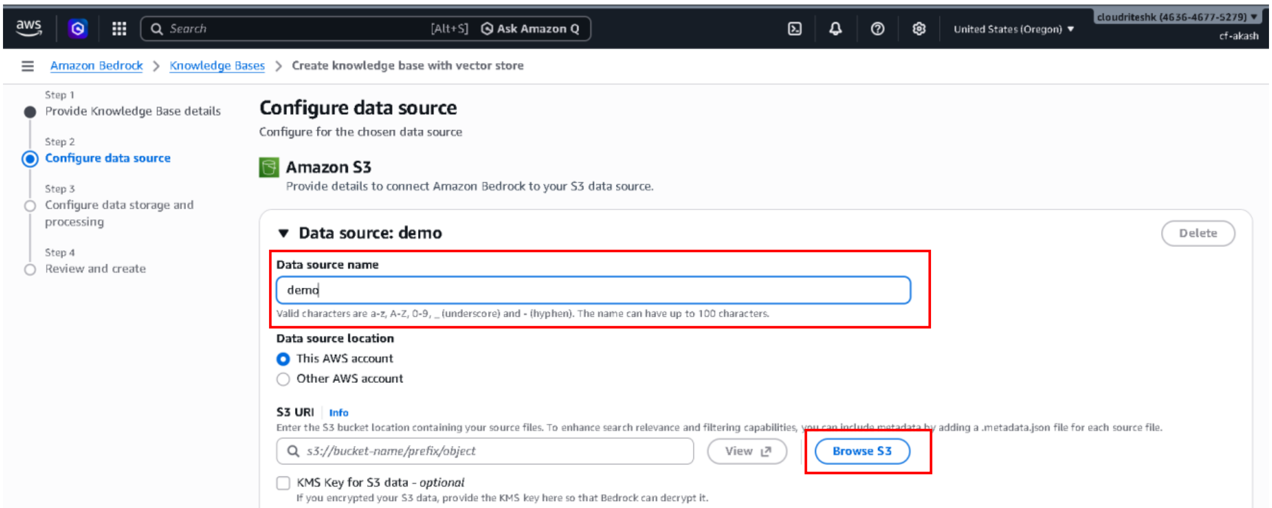
- Give Knowledge Base name
- Write Base description-optional
- IAM permission choose
    - ➢ Create and use a new service role.



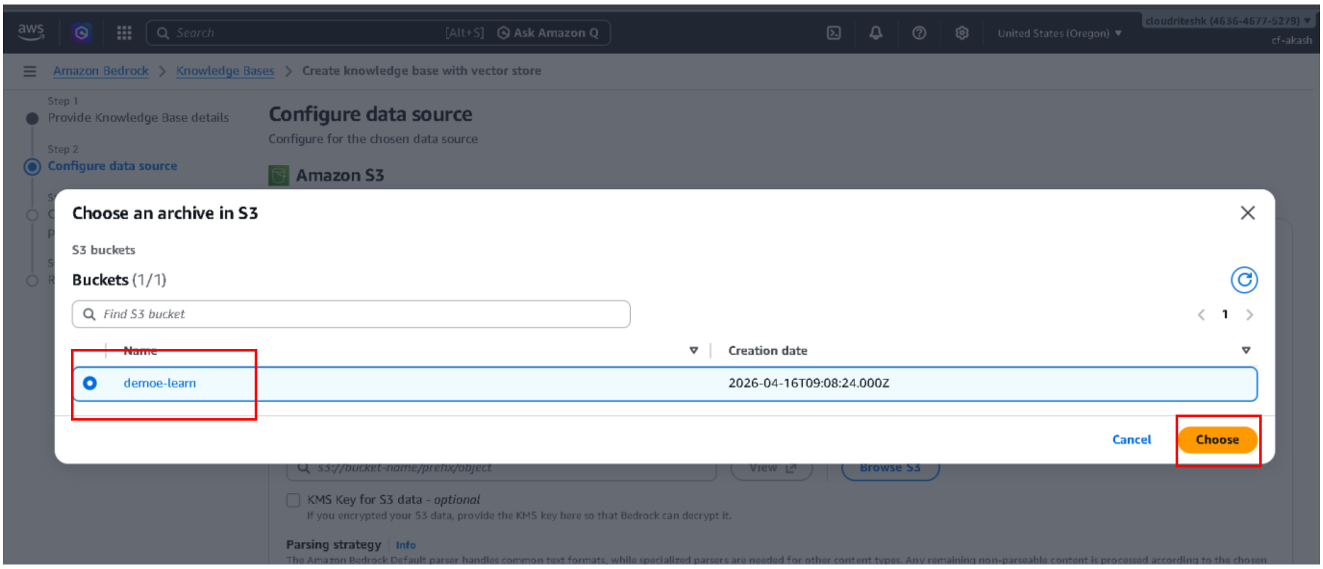
- Choose data source type:-
    - ➢ Amazon S3



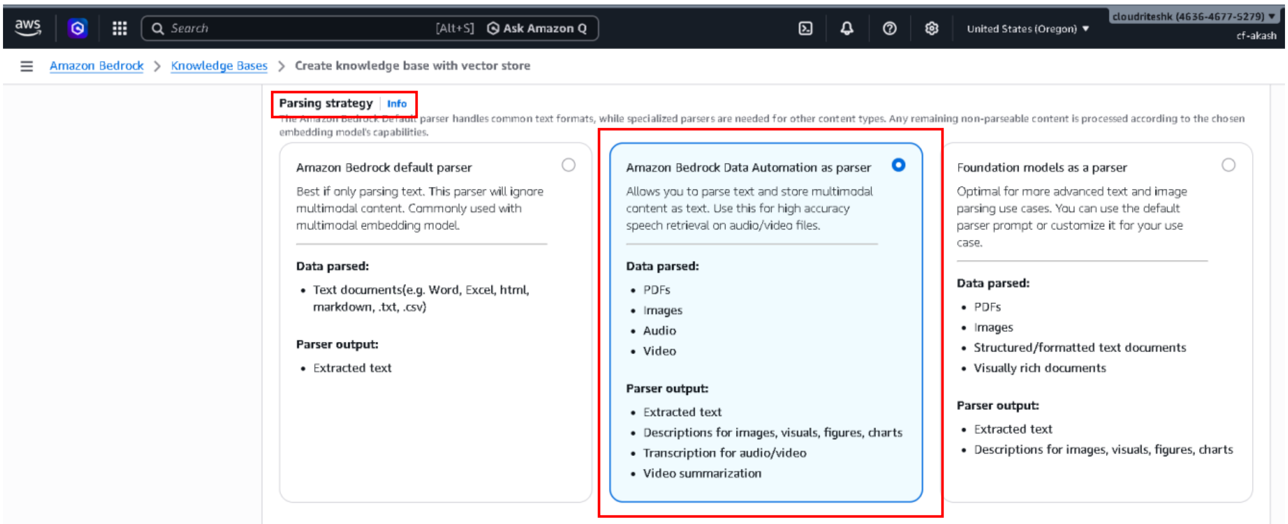
- Click Next.



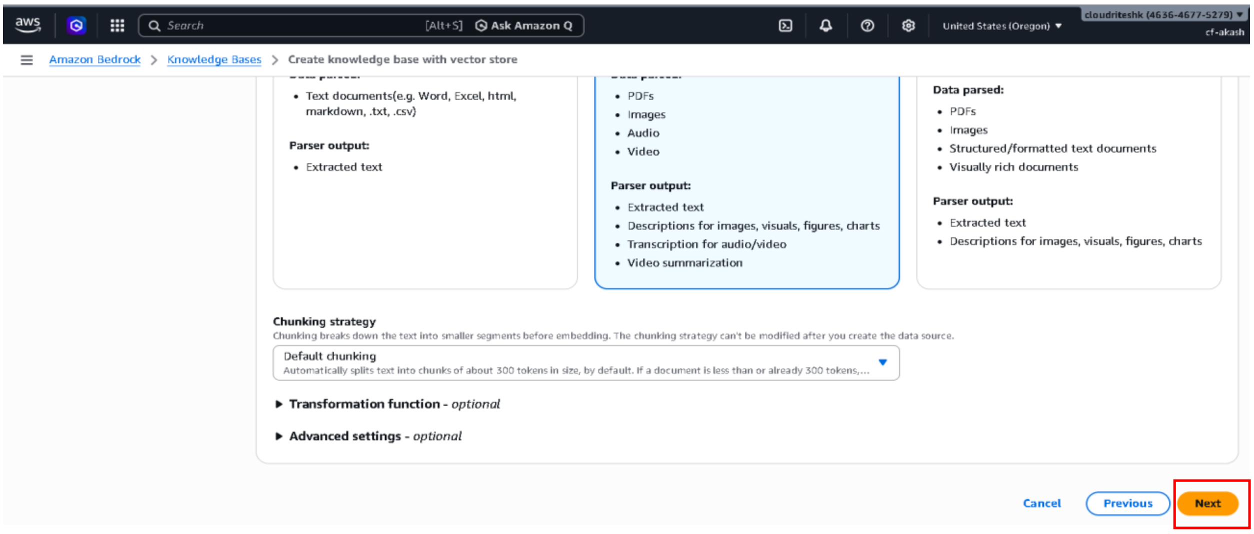
- Write data source name.
- Then Click on Browse S3.



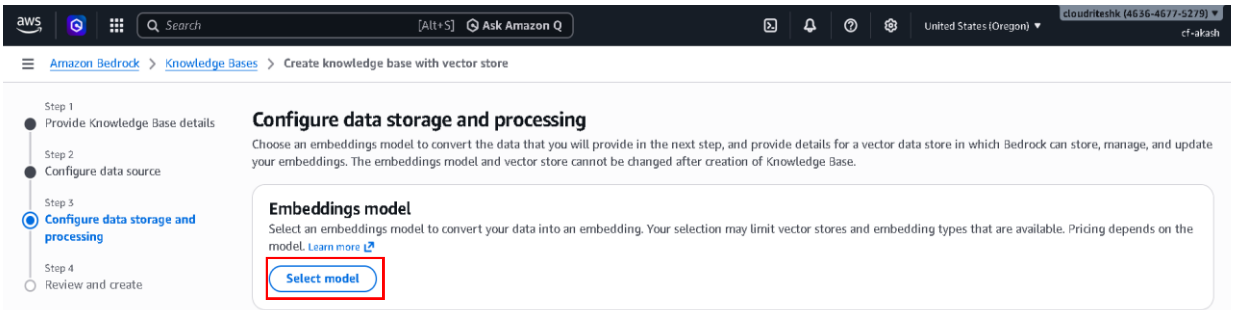
- Choose your created S3 bucket.
- And click on choose.



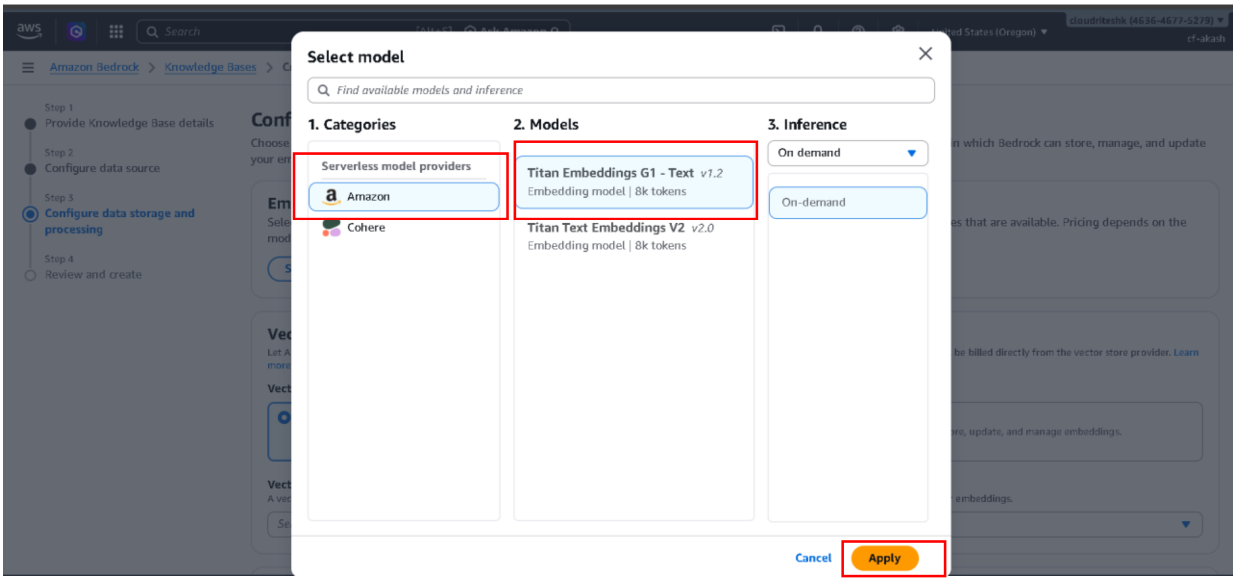
- Choose Parsing strategy
  - ➢ Amazon Bedrock Data Automation as parser



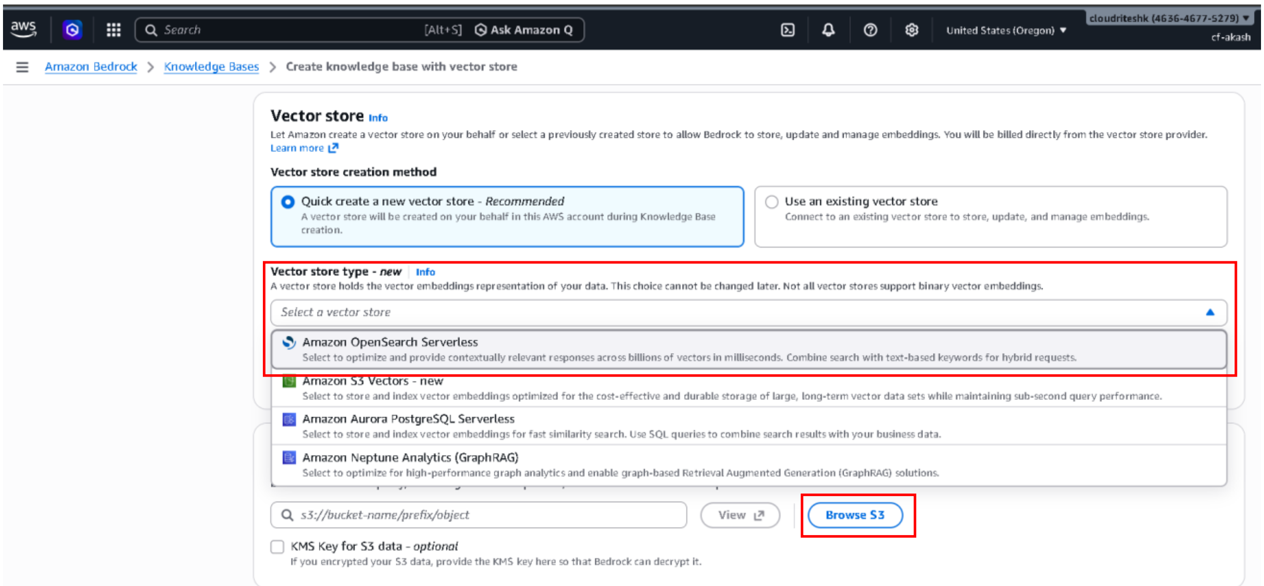
- Click on Next.



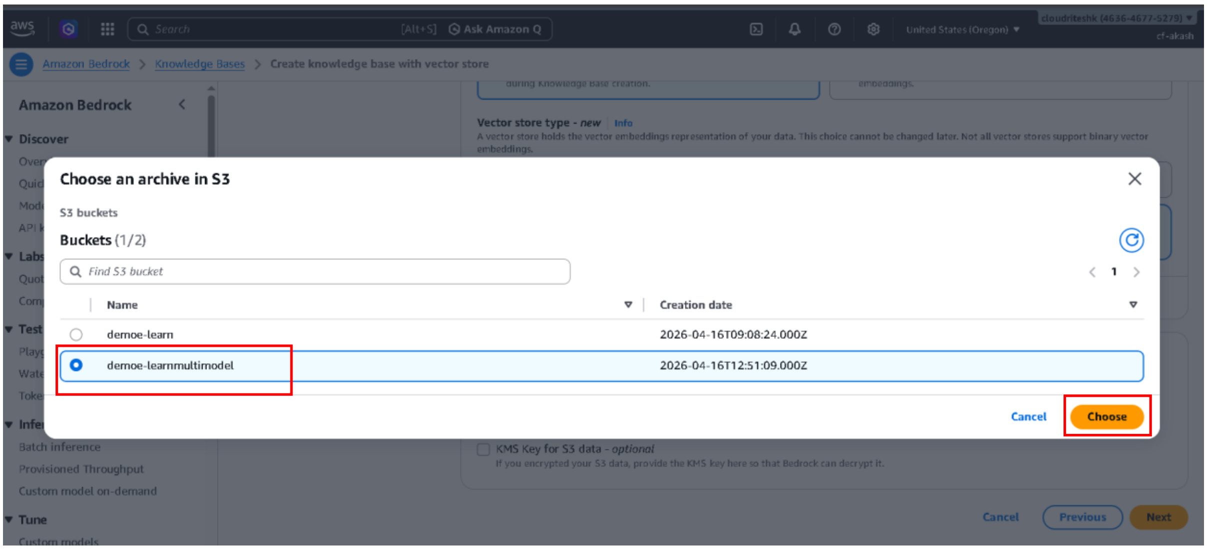
- Click on Select model



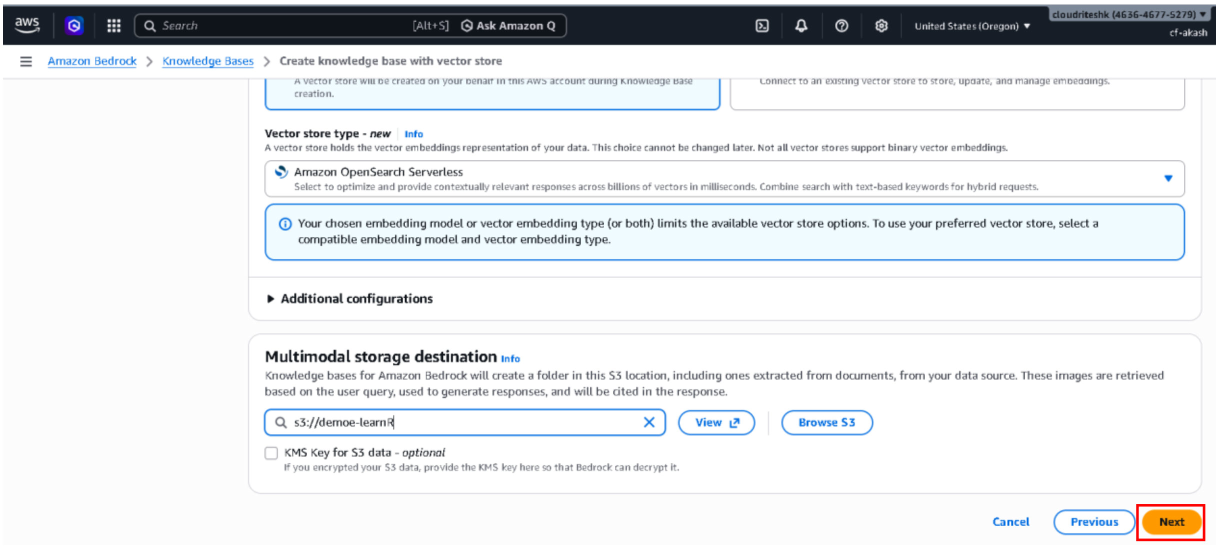
- Choose Amazon as service model provider
- Choose Model v1.2
- And now just click Apply.



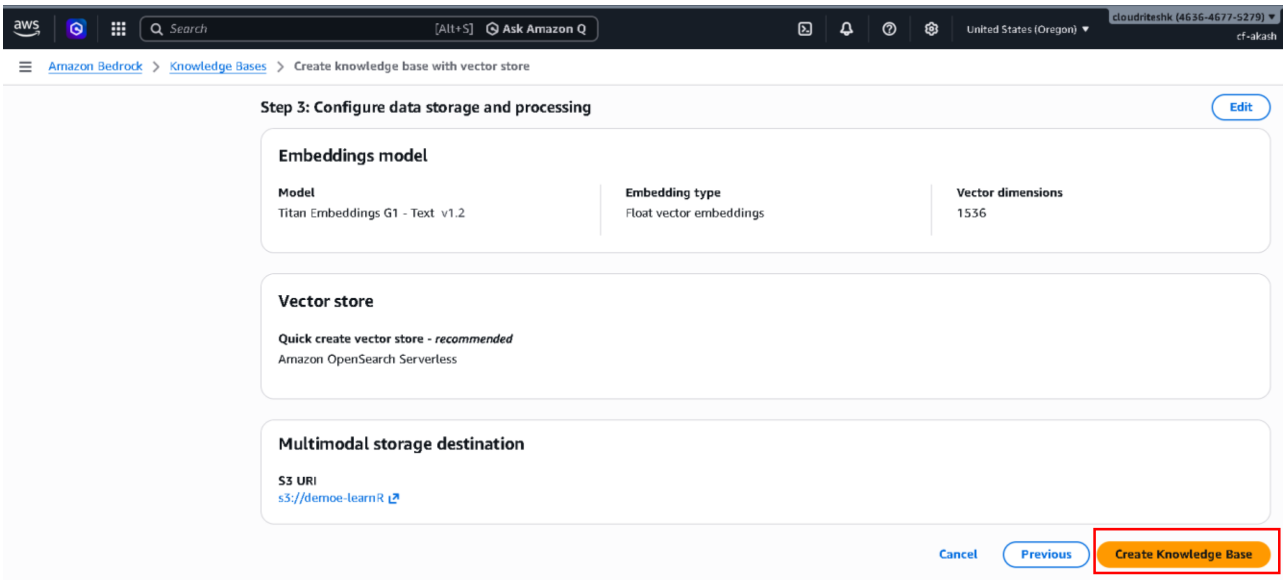
- Choose vector store type
  - ➢ Amazon OpenSearch Serverless.
- Click Browse S3.
- Now need one more s3 bucket to store images, video.
- So create new S3 bucket like we create in previous steps



- Select new S3 bucket and click choose

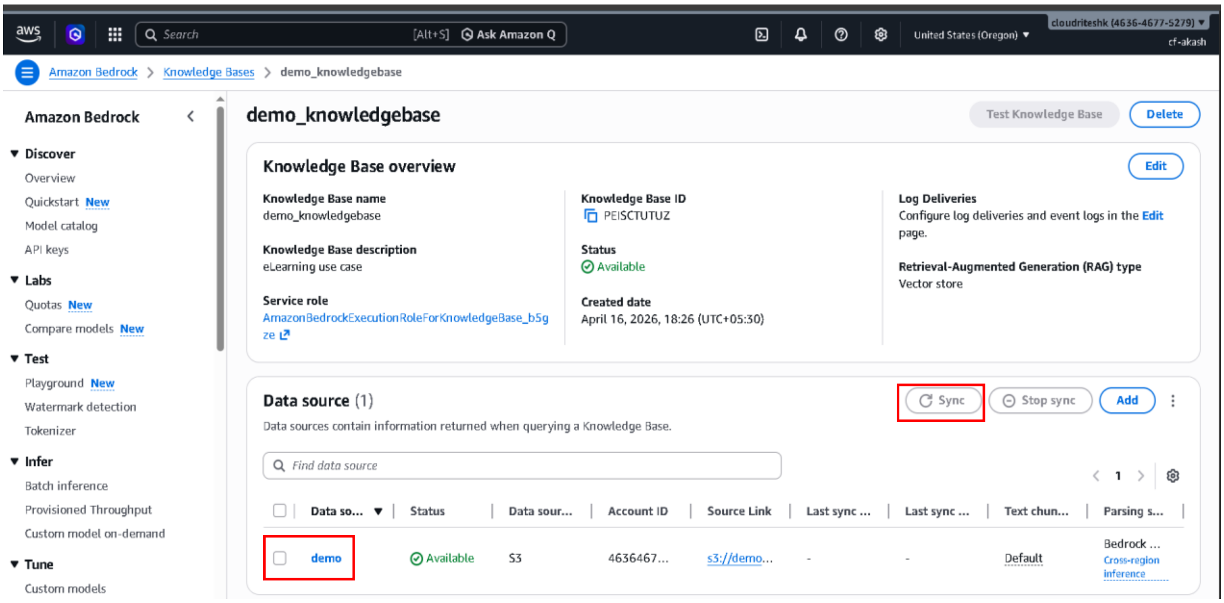


- Now click Next

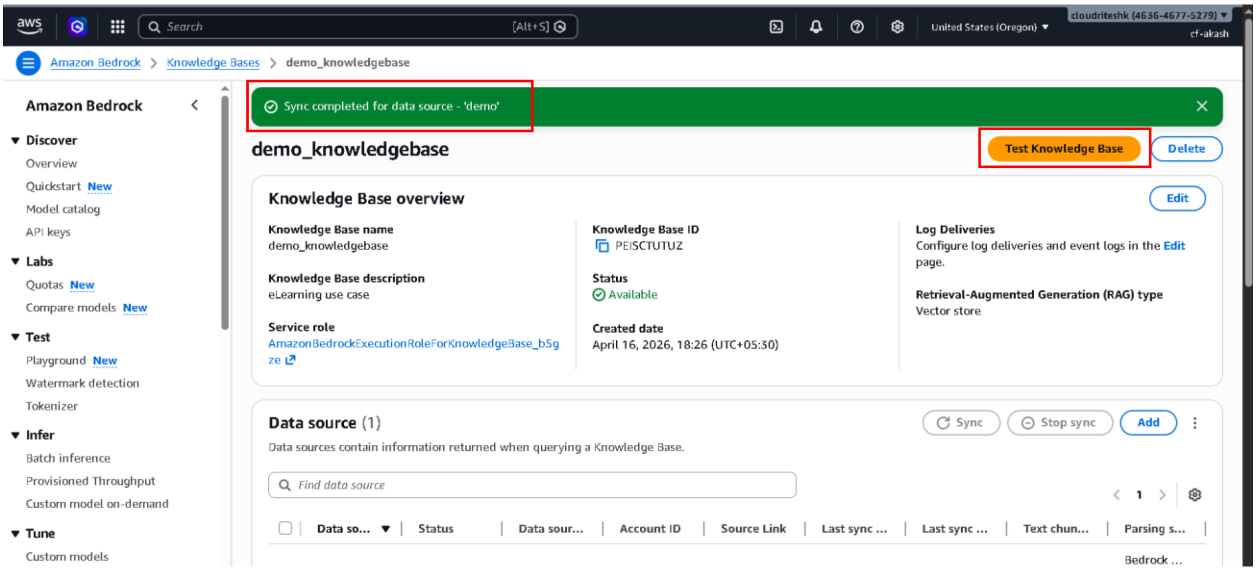


- Scroll down and click Create Knowledge Base.

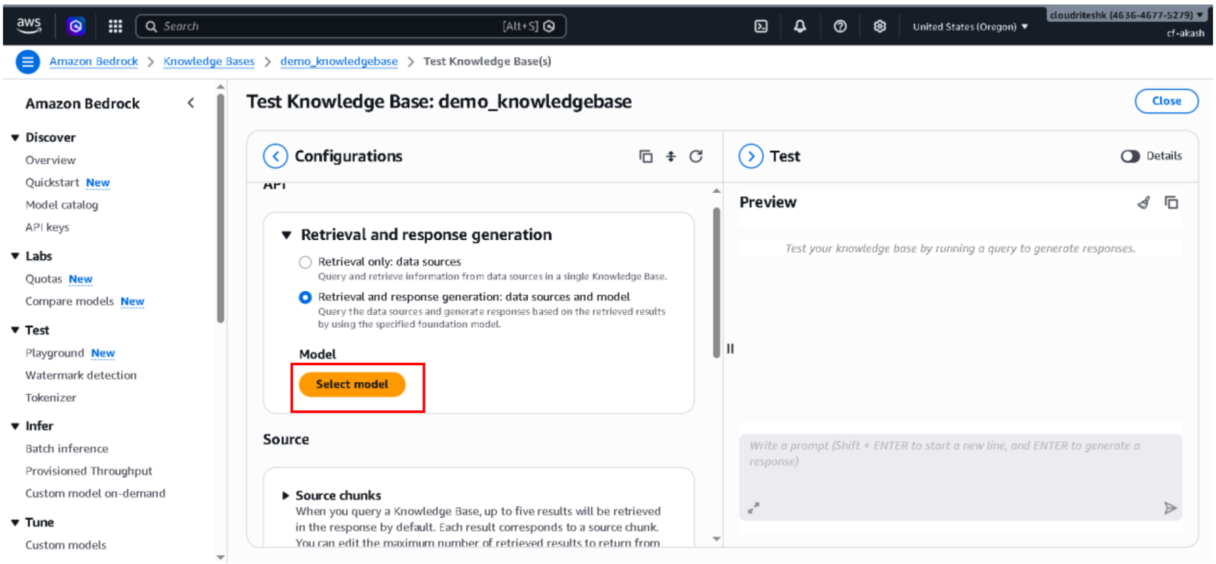
- It take’ 5 minute for creation.



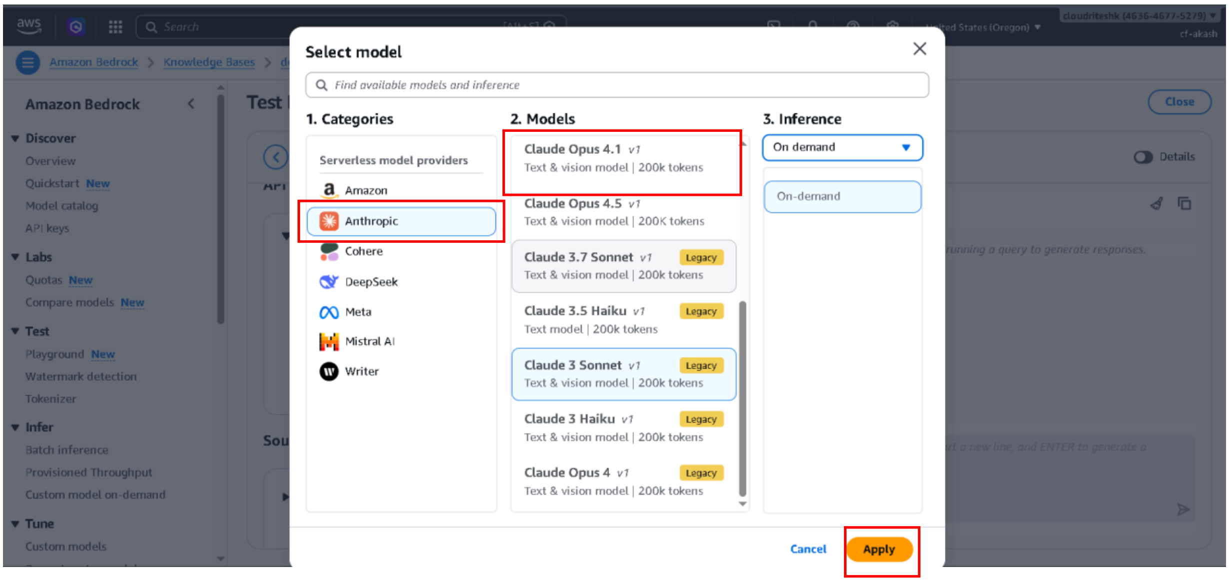
- Now our Knowledge Base is ready.
- Now just select Data source and click sync.
- Syncing time totally depends on your data size.



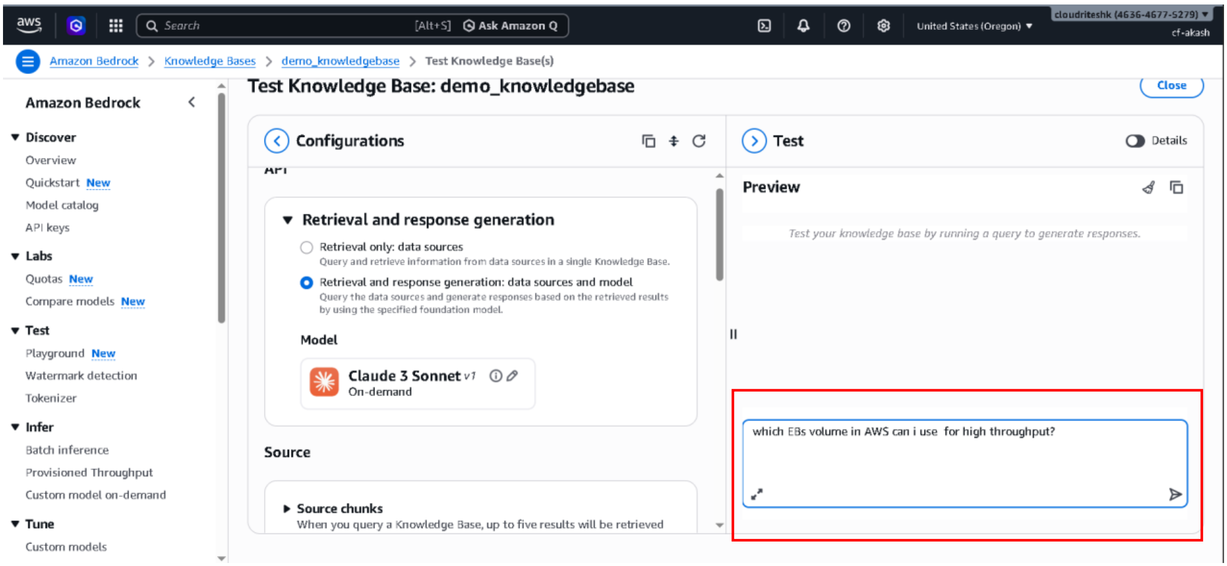
- Sync completed.
- Now just click Test Knowledge Base.



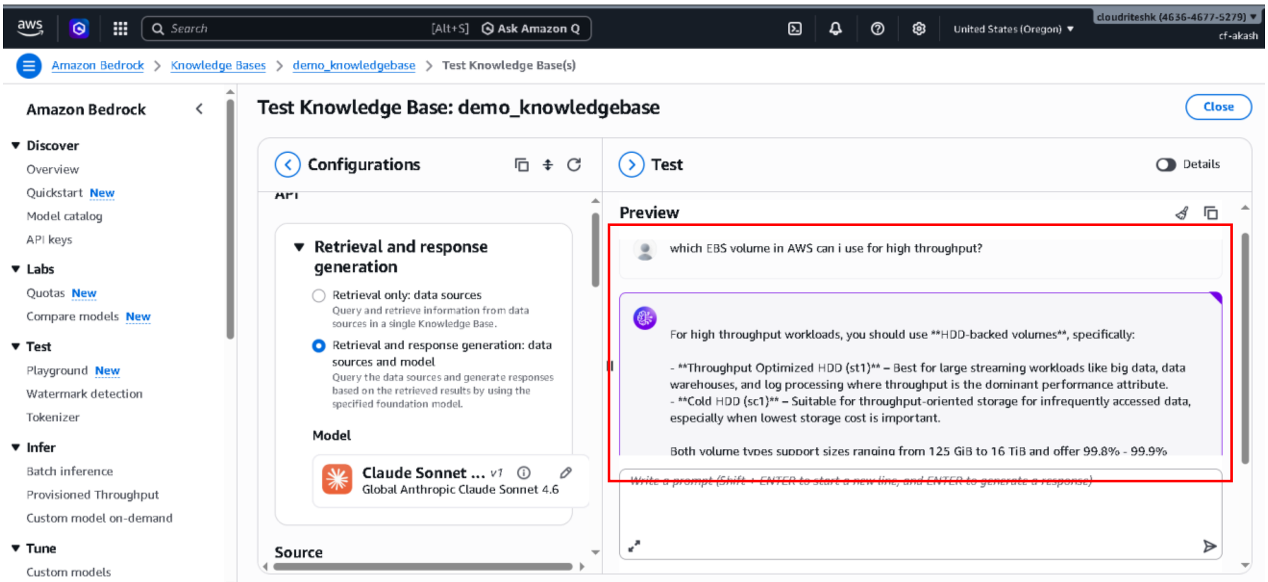
- Now Just click Select model.



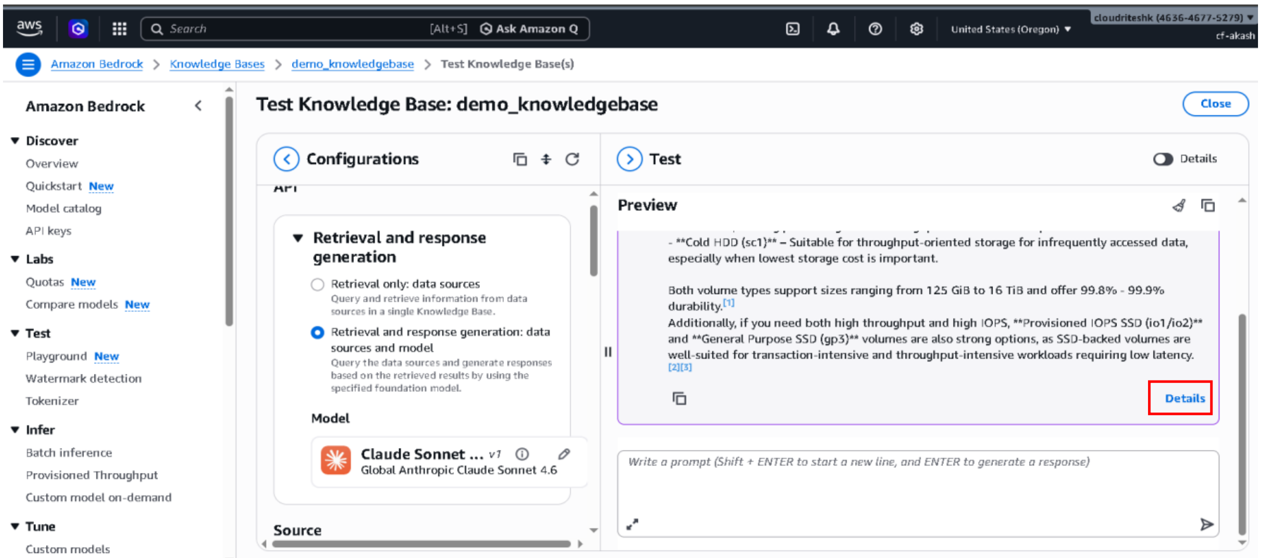
- Select categories and Model.
- And now simply just click Apply.
- A combination of this cloud foundation and data we have in our Knowledge Base is going to be able to respond to our Questions.
- Now ask some Questions.



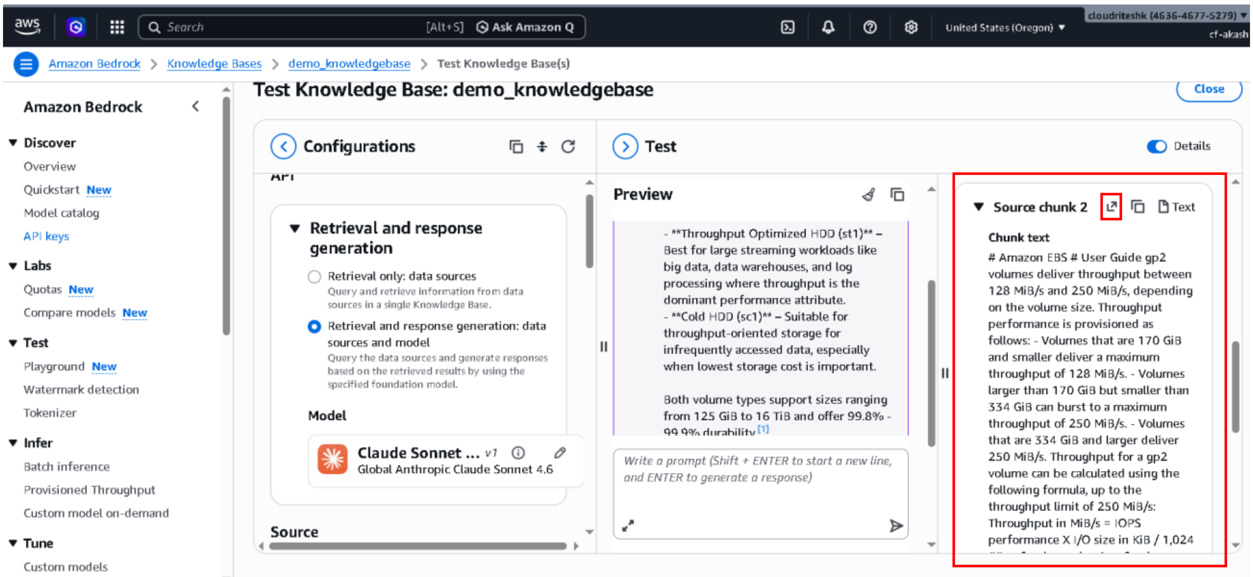
- Now this is the Question I Ask
- Now let's see what they can reply.



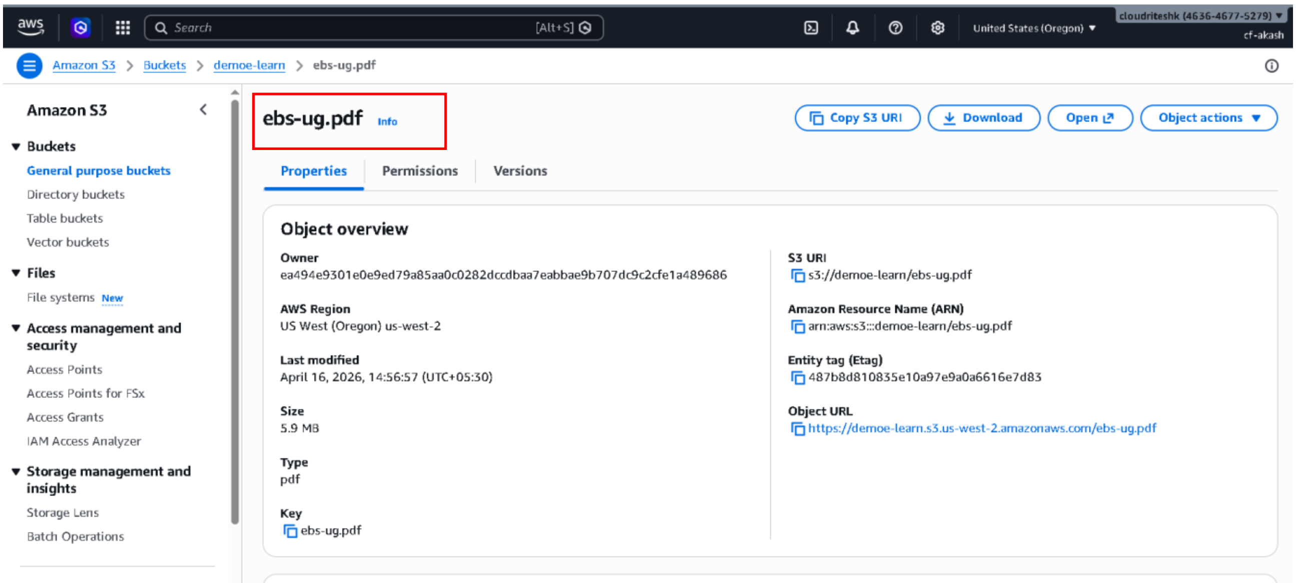
- It's working perfectly.
- Now scroll down.



- Click on Details.



- Here they show this response from this source that we have uploaded in S3 Bucket
- Now click on top first icon in source chunk 2



- So it is show you the source of this response
- So it is ebs-ug.pdf that we had uploaded as our source data.